INDEPENDENT AUDIT REPORT

Equity Research Audit

Three big-tech research agent skills evaluated against institutional research standards

Repository: github.com/nordeim/my-pi-agent @ 48cd3f97 (2026-10-30)

Scope: analyzing-us-bigtech-stocks · big-tech-earnings-postmortem · bigtech-equity-research

Evidence: 17 files · 2,078 lines · 3 executed scripts · 19 verification checks

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1. Executive Summary

This report presents an independent audit of three agent skills in the repository **nordeim/my-pi-agent** (cloned 2026-10-30, HEAD 48cd3f97, commit message “update skills”): **analyzing-us-bigtech-stocks**, **big-tech-earnings-postmortem**, and **bigtech-equity-research**. The evaluation question was fixed by the engagement brief: would this work meet the standards expected of major investment banks and institutional fund managers on Wall Street? To answer it, all 17 in-scope files were read in full, all three shipped scripts were executed against positive and negative test cases, and the skills' own embedded “verified” facts were tested against twelve external sources including SEC filings, Bloomberg, Reuters, WSJ and Yahoo Finance. Nineteen verification checks were recorded; every material claim in this report carries a confidence tag (Verified / Reasoned / Assumed).

The verdict: **none of the three skills passes a full bulge-bracket research-controls review as shipped, and one of them is genuinely close.** Skill 1, analyzing-us-bigtech-stocks, is the strongest artifact — its sourcing tiers, forced-hypothesis discipline, GAAP-adjusted bridge and dual executable audit gates are institutional in design, and its worked example survived external verification better than anything else audited: the GOOG close of $282.32 (−12.29%), the −$9.1B quarterly free cash flow, the $2.67 adjusted EPS, the $42.3B operating cash flow, the $30.7B Cloud result and the “second-largest one-day market-value wipeout ever” ranking were each confirmed against Bloomberg, Reuters, the earnings-call transcript or Yahoo Finance. Its gaps are an evadable numeric-audit matcher (H-4), coordinate assumptions in the render audit (M-1), and one cross-class figure that conflicts with a sibling skill (H-1).

Skill 2, big-tech-earnings-postmortem, contains the best single-file methodology prose of the three — its durability-versus-evidence driver classification and its failure-mode catalog (“sell-side ventriloquism”, the “accounting-noise hand-wave”) read like real desk experience — but it is the only skill that promises “institutional-grade” research deliverables while containing **no disclaimer language whatsoever** (H-2), and it ships zero enforcement artifacts: every quality gate is prose (M-5). Skill 3, bigtech-equity-research, contributes the most sophisticated epistemics in the repo — a Data-Gap Ledger and an anti-fabrication protocol for offline tooling — but its segment reference file misstates ASC 280 reportable segments for three of seven companies, verified directly against the FY2025/FY2026 10-Ks (H-3); its quant helper's “self-test” prints without asserting (M-4); and its worked example contradicts skill 1's on the same real event (H-1).

The deepest deficiency is portfolio-level. Three skills claim the same trigger space (“why did big tech X move after earnings”) with three different deliverable formats, two different

confidence taxonomies, and three mutually inconsistent doctrines on whether a price move may be allocated across drivers in percentage terms (M-2, M-3, M-6). The repository thus violates, across skills, the exact conflict-recording discipline the skills themselves enforce within a report. Twenty findings were classified in total; none is Critical in the security or data-loss sense, four are High, seven Medium, six Low and three Informational. A four-item P0 list — segment-map correction, disclaimer, matcher fix, worked-example reconciliation — would close most of the distance between skill 1 and the stated bar.

Table 1: Audit outcome summary

Severity	Count	Findings
Critical	0	None. No security vulnerability, data-loss path or production crash was found.
High	4	H-1 conflicting verified figures across skills · H-2 no disclaimer in skill 2 · H-3 segment-map ASC 280 errors · H-4 numeric-audit false-pass hole
Medium	7	M-1 render-audit coordinates · M-2 attribution doctrine conflict · M-3 confidence-taxonomy drift · M-4 non-asserting self-test and sign error · M-5 unenforced quality bar in skill 2 · M-6 routing overlap and catalog non-indexing · M-7 environment-coupled Playwright dependency
Low	6	L-1 packaging drift · L-2 no root LICENSE · L-3 formulas.md fence and circular pair · L-4 placeholder scan on raw HTML · L-5 phase/section mislabel · L-6 unverifiable embedded sell-side specifics
Informational	3	I-1 no MNPI/Reg FD statement · I-2 skill 3 lacks a GAAP bridge phase · I-3 consensus-vendor attribution variance

Scorecard averages on the ten-dimension institutional rubric (Section 5.3, anchored to the evidence in Sections 3 and 6): **skill 1 = 4.45 / 5**, **skill 2 = 2.85 / 5**, **skill 3 = 3.00 / 5**. The rubric scores are the auditor's reasoned judgments, not measurements; the anchors and per-dimension reasoning are stated alongside them.

2. Scope, Standards and Method

2.1 Scope and repository context

my-pi-agent is a customization bundle for the Pi agent (themes, extensions and a 340-skill library). The three audited skills sit under skills/ and total 17 files and 2,078 lines (Appendix A). Skill 1 ships a 238-line SKILL.md plus two JSON templates, three reference documents and two executable audit scripts. Skill 2 ships a single 283-line SKILL.md with no supporting artifacts of any kind. Skill 3 ships a deliberately lean 111-line SKILL.md plus four reference files, two templates and one Python helper. The audit

was read-only: the cloned repository was not modified, and all experiments ran from scratch copies under the auditor's own scripts directory.

Two repository-level facts frame the audit. First, the repository has **no root LICENSE file**; the only license declaration in scope is skill 2's frontmatter (Apache 2.0), which leaves the redistribution terms of skills 1 and 3 ambiguous (L-2). Second, the repository's own discovery layer — skills/skills-catalog.md and skills-inventory.md, which the README designates as the routing entry point — contains **no reference to any of the three skills** (case-insensitive search), so an agent routed through the catalog would never discover them (part of M-6). Both facts were verified directly in the clone.

2.2 The institutional standards applied

The evaluation rubric translates the engagement brief — “the highest standards expected of major investment banks and fund managers on Wall Street” — into seven concrete regulatory and craft frameworks, plus skill-engineering practice as a secondary axis. The table below lists each standard and where it bites in this audit. This is an analytical benchmark applied by analogy to agent skills, not a claim that the skills are regulated securities research; where a standard is inapplicable in form but applicable in substance (for example, FINRA 2241 as a disclosure-norm template for a non-broker artifact), that is stated.

Table 2: Standards framework and points of application

Standard	What it demands	Where it applies here
CFA Institute Code & Standards, V(A) and V(B)	Diligence and reasonable basis for every claim; communication distinguishes fact from opinion; limitations disclosed	Sourcing tiers, confidence labels, forced two-sidedness, hypothesis-exclusion discipline
SEC Reg S-K Item 10(e)	Non-GAAP measures reconciled to the most comparable GAAP measure; no greater prominence for non-GAAP	GAAP-vs-adjusted bridges, ex-item arithmetic on the page, consensus-basis statements
FINRA Rule 2241 (disclosure norms, by analogy)	Research disclosures: basis of conclusions, risk framing, no unconditional recommendations, disclaimer language	Disclaimers, scenario-not-price-target doctrine, risk/catalyst matrices
Regulation FD / MNPI discipline	Public information only; no material non-public information	Source tiers restricted to filings, press, transcripts; absence of an explicit MNPI statement noted (I-1)

Standard	What it demands	Where it applies here
OCC SR 11-7 model risk management (by analogy)	Conceptual soundness, testing, documentation, independent validation for quantitative code	Script self-tests, formula documentation, quant_helpers governance (M-4)
ASC 280 segment reporting	Use the issuer's actual reportable segments; treat the CODM measure with comparability caveats	segment-maps.md accuracy for seven companies (H-3)
Institutional research hygiene	Vintage dating of data, consensus-vendor attribution, error-correction policy, audit trail	Fact-base schema, append-only corrections, cutoff stamps, verification ledgers

2.3 Audit method and evidence classes

The audit proceeded in four passes. (1) **Full-file reads**: all 17 files were read end-to-end without truncation, plus the repository README, package.json, skills-catalog.md and skills-inventory.md for context. (2) **Script execution**: fact_base_audit.py was run in self-test, fact-base, checks and scan-only modes, then subjected to a purpose-built negative harness (four cases); quant_helpers.py was executed and every printed output compared with its documented target; overflow_audit.mjs was syntax-checked, usage-checked and executed against a purpose-built overflow probe page. (3) **External verification**: the skills' embedded “real, verified facts” about the Alphabet Q3 2026 event were checked against Bloomberg, Reuters, WSJ, Yahoo Finance and the earnings-call transcript, and the segment maps were checked against the NVDA FY2026 and AMZN FY2025 10-Ks on SEC EDGAR plus Apple's 10-K segment structure. (4) **Cross-skill comparison**: trigger descriptions, confidence taxonomies, attribution doctrines, deliverable formats and worked-example data were diffed across the three skills.

Every material claim below is tagged per the audit protocol: **Verified** means the check was executed and the result observed directly in this engagement; **Reasoned** means logical inference from verified evidence; **Assumed** means a stated assumption. Findings and fixes are kept separate — no repository code was rewritten during this audit, in accordance with the review-only scope.

2.4 Limitations

- GOOGL's official October 29, 2026 closing price could not be independently confirmed: Bloomberg's GOOGL quote page was robot-blocked, and the accessible Yahoo Finance prints (−12.78% / −12.79%, 283.35–283.37) match neither skill 1's −11.71% nor skill 3's −12.6%. Adjudication of that specific figure is therefore tagged Reasoned, while the existence of the cross-skill conflict is Verified.

- Analyst-action specifics embedded in skill 1's worked example (Citizens JMP downgrade, Wedbush price-target move, FactSet consensus figures, a BofA multiple comparison) could not be verified externally within this audit's time box. They are treated as unverified-by-audit, not as false — several surrounding figures in the same worked example did verify (L-6).
- overflow_audit.mjs was executed against a synthetic probe deck, not against a complete deck produced by the skill's own 15-page pipeline.
- The audit ran on 2026-10-30 against the clone at HEAD 48cd3f97; anything committed later is out of scope.

3. Findings by Severity

Findings are ordered by severity, not by file order. Each finding states location, description, evidence, impact, severity, a concrete recommended fix, and the auditor's confidence. Severity anchors: Critical blocks release (security, data loss, production crash); High is incorrect behavior or a material control gap on a common path; Medium is an edge-case defect, control weakness or consistency problem; Low is style, convention or documentation quality; Informational is advisory. No Critical findings were identified — an honest result, not a padded one: the scripts are well-engineered and the failure modes found are quality-assurance and consistency failures, not safety failures.

3.1 High-severity findings

H-1 · Cross-skill figure conflicts on the same event (High)

Table 3: Finding H-1 — Conflicting “verified” figures for the same event across skills — no single source of truth

Field	Content
Location	analyzing-us-bigtech-stocks/references/methodology.md §§3–4 (worked example) vs bigtech-equity-research/reference/formulas.md (worked examples)
Description	Both skills use Alphabet's Q3 2026 print (reported 2026-10-28) as their worked example, and both present their figures as verified — yet they publish materially different numbers for the same datapoints: the GOOGL move is −11.71% in skill 1 and −12.6% in skill 3; consensus free cash flow is −$3.5B in skill 1 and −$2.70B in skill 3; revenue is $127.57B against $128.2B consensus in skill 1 and $127.58B against $128.22B in skill 3.

Field	Content
Evidence	Direct file comparison (Verified). External checks: Bloomberg's GOOG quote page confirms skill 1's Class C figures exactly — $282.32 close, −$39.57, −12.29%, as of 5:20 PM EDT 2026-10-29 — implying the $321.89 prior close it claims; Yahoo Finance pre-event pages confirm the October 28 closes ($321.89 GOOG / $324.89 GOOGL). For GOOGL on October 29, accessible Yahoo prints show −12.78%/−12.79% (283.35–283.37), matching neither skill. Skill 1's implied GOOGL close is 286.85; skill 3's is 283.95. On Alphabet's ≈$3.95 trillion market value, the 0.89-point gap between the two skills' GOOGL moves is ≈$35 billion of implied value — the two skills disagree by more than the market capitalization of most S&P 500 constituents.
Impact	Two skills in one library can issue reports carrying materially different “Verified” figures for the same event; a portfolio manager or research-controls function cross-checking them would flag the desk. The repository violates, across skills, the conflict rule both skills enforce within a report — skill 1: “silent picking is forbidden”; skill 3: “never present a figure reconstructed from memory.”
Recommended fix	Create one canonical, dated, fully sourced worked-example file (or shared fact-base JSON) consumed by all three skills; add a repository-level consistency test that diffs shared datapoints across skills; reconcile the GOOGL figure against an exchange close record and record the correction append-only, exactly as the skills' own doctrine requires.
Confidence	Verified (the conflict and its magnitude); Reasoned (which GOOGL figure is correct — GOOG verified exactly, GOOGL adjudication inconclusive).

H-2 · No disclaimer anywhere in skill 2 (High)

Table 4: Finding H-2 — Skill 2 produces “institutional-grade” research deliverables with no disclaimer anywhere

Field	Content
Location	big-tech-earnings-postmortem/SKILL.md — the entire 283-line file
Description	The skill's stated ambition is a research artifact “a portfolio manager could defend in an investment committee.” The file contains zero occurrences of “investment advice”, “disclaimer” or “solicitation”. The quality-gates checklist omits any disclaimer requirement, and the deliverable spec (Step 9) bans placeholder values and fabricated citations but not missing disclosures.

Field	Content
Evidence	Case-insensitive grep across all three skills (Verified): skills 1 and 3 carry disclaimers in multiple places — skill 1's Definition of Done (“Disclaimer present: informational only, not investment advice”), its methodology §14 (“Informational only — not investment advice, not a solicitation. No price targets published.”) and its slide templates; skill 3's description, hard rule 7 and a dedicated Disclaimer section in sourcing-and-confidence.md. Skill 2 returns no matches in any form.
Impact	Measured against the FINRA 2241-style disclosure norms this audit applies by analogy, distributing an institutional-grade research artifact with no disclaimer or basis language is a compliance non-starter. It is also internally inconsistent: two sibling skills in the same repository treat the disclaimer as non-negotiable, and a publicly distributed skill teaches the omission by example.
Recommended fix	Add the disclaimer to the quality gates and to Step 9's design constraints, aligned with skills 1 and 3; while editing, add the public-sources-only statement (I-1). Estimated effort: minutes.
Confidence	Verified.

H-3 · Segment map wrong for 3 of 7 companies (High)

Table 5: Finding H-3 — Segment map misstates ASC 280 reportable segments for three of seven companies

Field	Content
Location	bigtech-equity-research/reference/segment-maps.md — the “Reportable segments” column
Description	The file opens with “Use the EDGAR 10-Q segment table as the source of truth,” then lists, as reportable segments: for Nvidia — Data Center, Gaming, Professional Visualization, Automotive, OEM (these are market platforms, i.e. revenue disaggregation inside Nvidia's two reportable segments); for Amazon — North America, International, AWS and Advertising (Advertising is a net-sales disclosure line, not a reportable segment); for Apple — iPhone, Mac, iPad, Wearables/Home/Accessories, Services (product-line disaggregation; Apple's reportable segments are geographic). The Google, Microsoft, Meta and Tesla mappings are correct.

Field	Content
Evidence	SEC EDGAR, retrieved during this audit (Verified). NVIDIA FY2026 10-K (nvda-20260125.htm): “The Compute & Networking segment includes our Data Center … and Automotive platforms … The Graphics segment includes GeForce GPUs for gaming and PCs, and Quadro/NVIDIA RTX GPUs …” — exactly two reportable segments, with Data Center and Automotive inside Compute & Networking. Amazon FY2025 10-K (amzn-20251231.htm): segment tables for North America, International and AWS only, with advertising discussed as a net-sales driver, not a segment. Apple's 10-K reports five geographic segments (Americas, Europe, Greater China, Japan, Rest of Asia Pacific; FY2025 net sales $178.4B / $111.0B / $64.4B respectively), with the product categories appearing as revenue disaggregation.
Impact	A Phase 3 “segment variance table” built from this map would attribute consensus variance to segments that do not exist as such in the filings — the exact class of error an institutional reader catches immediately, and the exact hallucination the skill's own sourcing rules exist to prevent. It also contradicts the file's own source-of-truth claim.
Recommended fix	Restructure the map into two columns — “Reportable segments (ASC 280)” versus “Key disclosed revenue disaggregation / market platforms” — per company, each with an as-of vintage (“per FY2026 10-K”) and a stated refresh cadence; add the comparability caveat that segment operating income is the CODM-reviewed measure and is not comparable across issuers.
Confidence	Verified (SEC filings retrieved and quoted).

H-4 · Numeric audit gate can silently pass (High)

Table 6: Finding H-4 — The numeric audit gate can silently pass absent numbers (unanchored substring matching)

Field	Content
Location	analyzing-us-bigtech-stocks/scripts/fact_base_audit.py — token_in(), lines 59–66
Description	Fact tokens are matched against slide text with unanchored substring tests, so a fact's digit string “passes” whenever it appears inside any other number printed on the page. The docstring's sign/symbol tolerance is a deliberate and reasonable design goal, but the implementation lacks digit-boundary anchoring.

Field	Content
Evidence	Executed harness, built and run for this audit (Verified): a fact recording a close of 82.3 against a slide printing only “$282.32” returns “PASS — '82.3' present” and exit code 0 (CLEAN); a fact of 12.29 against a slide printing only “−112.29%” likewise returns PASS and exit 0. Control cases behave correctly: a genuinely present number passes; a genuinely absent number (63.4) fails with exit 1. The script's own --selftest passes; these evasion cases are not in it.
Impact	Skill 1's golden rule 5 — “a report that has not passed the numeric and render audits is not finished” — is weaker than claimed. Number drift from the fact base can ship undetected whenever digits overlap, a realistic occurrence with prices and percentages that share digits (82.3 inside 282.32 is a live example, not a contrivance). In Wall Street QA terms, this is a control with a known evasion path.
Recommended fix	Match tokens on digit boundaries — for example a regex of the form (?<![\d.])TOKEN(?![\d.]) applied to the symbol-stripped text, or tokenize the slide's numbers and compare parsed values under a formatting-normalization table; add the two evasion cases above to --selftest so the gate proves its own anchoring.
Confidence	Verified (executed and observed).

3.2 Medium-severity findings

M-1 · Render audit measures viewport, not slide origin (Medium)

Table 7: Finding M-1 — Render audit measures against the viewport, not the slide origin; no mandated body reset

Field	Content
Location	overflow_audit.mjs evaluate() (lines 85–105); report-design.md §1 (design tokens omit a html,body margin reset)
Description	The overflow check compares each element's bounding rectangle to the viewport dimensions (W×H), not to the .slide element's own box. Any deck whose page omits a body margin reset — the design tokens never mandate one — offsets the entire 1280×720 canvas by the default 8px, and every full-width element then reports a right edge at 1288px and fails the 2px-tolerance check.
Evidence	Probe execution (Verified): a purpose-built slide with a genuinely overflowing element and a properly sized .slide produced “OVERFLOW … clipped: h1 bottom=66 right=1288 | p bottom=106 right=1288 | div.too-low bottom=755” — the h1 and p flags are pure body-margin artifacts; only div.too-low was a true overflow. A secondary observation: the page load uses waitUntil networkidle against a design system that @imports Google Fonts, which can stall toward the 30-second timeout in offline environments.

Field	Content
Impact	Systematic false positives train users to dismiss OVERFLOW flags — the skill's own protocol then normalizes “documented false positives” — or, if a generator happens to reset margins, genuine 8px-scale defects are measured against a shifted origin. Either way the gate's authority erodes.
Recommended fix	Compute overflow relative to the .slide element's bounding rect (offset-aware), or mandate html,body{margin:0;padding:0} in the token sheet and audit for it; make page loading fall back from networkidle to domcontentloaded with a bounded wait when fonts do not resolve.
Confidence	Verified (probe executed).

M-2 · Three inconsistent attribution doctrines (Medium)

Table 8: Finding M-2 — Three mutually inconsistent doctrines on attribution quantification

Field	Content
Location	Skill 1 methodology §9; skill 2 SKILL.md “Common failure modes” and Step 3; skill 3 SKILL.md Definition of Done and methodology Phase 7
Description	On the core methodological question — may the analyst allocate a realized price move across drivers in percentage terms — skill 1 says conditionally (“ordinal ranks only. Publish a percentage attribution only if it is computable from sourced data; otherwise write attribution is qualitative”), skill 2 says never (“False precision: inventing a percentage of the move attributable to each driver. Use ordinal ranks, not fake decompositions”), and skill 3 says always, as a Definition-of-Done item (“Waterfall quantified, sums to the realized move”), softened only by its hard rule that weights are Reasoned.
Evidence	Direct quotation from the three files (Verified). Skill 3's own worked waterfall allocates −12.6 points across beta, FCF/capex, EPS quality, Search and positioning; skill 2's failure-mode list would classify exactly that artifact as a “fake decomposition” if the weights are not regression-derived.
Impact	Skill routing silently decides the answer to a methodological controversy: the same question yields a ranked table from one skill and a quantified waterfall from another, and the outputs are not comparable. A research desk operates one attribution policy, not three.
Recommended fix	Adopt skill 1's conditional rule as the repository canon — quantify the market-wide beta component (computable from sourced index returns), label idiosyncratic splits explicitly as Reasoned analytical allocations, and never present weights as regression output — then make skills 2 and 3 reference the shared policy instead of restating divergent ones.

Field	Content
Confidence	Verified (textual).

M-3 · Confidence taxonomy drift across skills (Medium)

Table 9: Finding M-3 — Confidence-taxonomy drift across skills, and an internal contradiction inside skill 3

Field	Content
Location	Skill 1 SKILL.md “The confidence system”; skill 2 Step 7; skill 3 sourcing-and-confidence.md and methodology.md Phase 1
Description	The three skills ship different confidence taxonomies with different evidentiary bars for the same labels. Skill 1 uses three labels (Verified / Reasoned / Assumed) with Verified = observed in ≥1 named primary source and decision-critical items cross-checked in ≥2. Skill 2 uses four (adding Unverifiable) with Verified = “executed/checked and observed directly.” Skill 3 uses four with Verified = “≥2 independent primary/near-primary sources, consistent to the reported precision.” Separately, skill 3 contradicts itself on transcript providers: methodology.md Phase 1 lists Motley Fool / Seeking Alpha transcripts as source preference 2 (cross-checked against audio), while sourcing-and-confidence.md ranks Motley Fool in tier 6 aggregators (“verify the underlying number elsewhere”); skill 1 ranks the same transcripts T1.
Evidence	Direct quotation from the four files (Verified).
Impact	The same badge asserts different evidentiary bars depending on which skill fired, so outputs are not comparable and a quality function cannot apply one standard; the “Verified” guarantee degrades from a specification to an inclination. The intra-skill-3 contradiction additionally leaves transcript tiering undefined at exactly the point where earnings-postmortem evidence is densest.
Recommended fix	Extract one shared sourcing-and-confidence reference (single taxonomy, single source hierarchy with transcript-provider tiering resolved) and have all three SKILL.md files consume it; keep the stronger of duplicate definitions and delete the weaker rather than reconciling by paraphrase.
Confidence	Verified (textual).

M-4 · Self-test prints without asserting; sign error (Medium)

Table 10: Finding M-4 — quant_helpers self-test does not assert; formulas.md carries a sign error; no unit tests

Field	Content
Location	bigtech-equity-research/scripts/quant_helpers.py __main__ block; reference/formulas.md “OCF shortfall” worked example
Description	The __main__ block prints computed values next to hand-written comments but contains no assertions, so a silent formula regression would still print a healthy-looking “self-test.” The OCF-shortfall worked example in formulas.md states “(−2.70 + 55.09) − 42.29 = −10.10B”, while the same arithmetic executed in the script returns +10.10B — the document and the code disagree on the sign convention of the skill's signature derived metric. avg_pt_cut has no zero-division guard, and the module ships with no unit tests of any kind.
Evidence	Executed (Verified): python3 quant_helpers.py reproduces every documented target — revenue delta −0.64B (−0.50%), EPS delta −0.90 (−29.80%), alpha −11.1pp, capex/OCF 121.4%, average PT cut −13.7%, waterfall sum 12.6 — and prints “OCF shortfall: +10.10B” against formulas.md's −10.10B. The arithmetic identity is unambiguous: (−2.70 + 55.09) − 42.29 = +10.10.
Impact	Under the SR 11-7 analogy this is untested model code inside a research pipeline: regressions ship silently, and a junior analyst following formulas.md states a negative shortfall (or the sign silently flips between documentation and tool). The other skill's audit script demonstrates the correct pattern — an asserting self-test — so the repo already contains the fix as a template.
Recommended fix	Convert the __main__ block to an asserting self-test in the fact_base_audit.py style; correct the formulas.md example or define the sign convention in both places in one sentence; add pytest coverage including consensus=0, old_pt=0, empty inputs and rounding boundaries.
Confidence	Verified (executed).

M-5 · Skill 2 quality bar is unenforced prose (Medium)

Table 11: Finding M-5 — Skill 2's quality bar is unenforced prose — no gates, templates or references ship

Field	Content
Location	big-tech-earnings-postmortem/ — a single SKILL.md, no scripts/, templates/ or references/

Field	Content
Description	Skill 2's quality gates — “FCF is computed, not copied”, “a verification ledger exists with 10+ load-bearing claims”, “no fabricated numbers” — have no executable counterpart, unlike skill 1's two audit scripts, and not even a static template: skill 3 ships an evidence-ledger template that skill 2 reimplements in prose.
Evidence	File-tree verification (Verified): one file, 283 lines, zero supporting artifacts.
Impact	The same user request routed to skill 2 yields a deliverable with materially weaker guarantees than routing to skill 1, for identical inputs. The gap between claimed rigor and enforceable rigor is exactly what a research-controls function measures.
Recommended fix	Either port the fact-base and audit-gate pattern into skill 2 (or have its Phase- equivalent explicitly invoke skill 1's scripts), or demote skill 2 in its own description to a lightweight fast path with the limitations stated; reuse skill 3's evidence-ledger template rather than restating it.
Confidence	Verified (file tree).

M-6 · Routing overlap; catalog indexes none (Medium)

Table 12: Finding M-6 — Routing overlap: three skills claim the same trigger space, and the catalog indexes none of them

Field	Content
Location	The three SKILL.md descriptions; skills/skills-catalog.md and skills-inventory.md (repository discovery layer)
Description	All three descriptions trigger on variants of “why did big-tech X move after earnings”: skill 1 (“Use when asked why a stock moved, to run a root-cause analysis (RCA) on a price event”), skill 2 (“Use when the user asks why a big-tech stock moved sharply after earnings”), skill 3 (“Use when asked to explain a stock-price event (post-earnings move, guidance shock, regulatory action)”). None of the three references the others. Meanwhile the repository's catalog — which the README designates as the routing entry point — contains no entry for any of the three skills (case-insensitive search over both catalog files).
Evidence	Frontmatter text and catalog greps (Verified).
Impact	Skill selection for the highest-value research request in the library is nondeterministic, and the catalog routing layer would never surface the skills at all. Three deliverable formats (15-page deck, 7-section HTML, 13-section note) for one question family multiplies maintenance surface — which is precisely where the drift in M-2, M-3 and H-1 already shows.

Field	Content
Recommended fix	Demarcate the three in their descriptions (deck-with-audit-gates vs lightweight post-mortem vs quantified-note), cross-reference each other explicitly, and register all three in skills-catalog.md with a note that the catalog must be regenerated when skills land.
Confidence	Verified.

M-7 · Implicit, environment-coupled Playwright dependency (Medium)

Table 13: Finding M-7 — Playwright dependency is implicit and environment-coupled

Field	Content
Location	overflow_audit.mjs loadPlaywright() — fallback resolution bases include the hardcoded absolute paths /home/z/my-project and /home/z/my-project/scripts; repository package.json (no playwright); SKILL.md Phase 10 (prerequisite unstated)
Description	The render audit resolves Playwright from the working directory, the script's own directory, or two machine-specific absolute paths. Playwright is absent from the repository's dependencies, and the skill text never states the install prerequisite — the script merely prints an install hint after failing.
Evidence	Code inspection plus execution (Verified): the script ran in this audit environment only because the auditor's sandbox happens to match one of the two hardcoded paths; the repository itself has no node_modules and no playwright dependency entry.
Impact	The Definition-of-Done line “both audit scripts exit clean” is unverifiable in a fresh clone, and shipped skill code contains author-machine absolute paths — a portability smell that will surface as an environment error (exit 2) for most users on first use.
Recommended fix	State the prerequisite in SKILL.md Phase 10 and the Bundled-resources table (npm install playwright && npx playwright install chromium); resolve from the working directory and a documented relative location only; remove the absolute paths.
Confidence	Verified (code read; behavior observed).

3.3 Low-severity findings

L-1 · Packaging and frontmatter drift (Low)

Table 14: Finding L-1 — Packaging and frontmatter drift across the three skills

Field	Content
Location	Folder structure and YAML frontmatter of the three skills
Description	Skill 1 uses references/ + assets/ + scripts/; skill 3 uses reference/ + templates/ + scripts/; skill 2 ships no subdirectories. Frontmatter keys are inconsistent: skill 1 carries name and description only; skill 2 adds license and a non-standard compatibility key; skill 3 adds version — none matches the repository spec's full optional set (license / author / version), and only skill 2 declares any license.
Evidence	Directory listing and frontmatter parsing (Verified): description lengths 602 / 607 / 539 characters — all within the 1024-character limit, and all three names match their folders, so spec compliance in the narrow sense holds.
Impact	Discoverability and tooling assumptions suffer; the license asymmetry feeds L-2.
Recommended fix	Standardize on one folder convention and one full frontmatter set across the three skills.
Confidence	Verified.

L-2 · No root LICENSE (Low)

Table 15: Finding L-2 — No root LICENSE in a 340-skill public repository

Field	Content
Location	Repository root (no LICENSE file); skill 2 frontmatter (Apache 2.0)
Description	The repository ships skills publicly with no top-level license; the only declaration in scope is skill 2's Apache 2.0, which does not cover skills 1 and 3 whose content includes substantial original methodology and code.
Evidence	Filesystem check (Verified).
Impact	Redistribution terms for the audited skills are ambiguous — a licensing reviewer would flag it before any institutional reuse of the methodology.
Recommended fix	Add a root LICENSE and per-skill license fields; state whether the Apache 2.0 declaration on skill 2 was intended repo-wide.
Confidence	Verified.

L-3 · Markdown defect; circular market-cap pair (Low)

Table 16: Finding L-3 — formulas.md markdown defect and a circular market-cap pair

Field	Content
Location	bigtech-equity-research/reference/formulas.md, “Market cap” section (lines 66–73)
Description	A stray closing code fence causes the trailing sentence (“Run the math with scripts/quant_helpers.py to avoid arithmetic errors.”) to render inside a code block. The section's two formulas are also mutual inverses (market-cap-erased = pre-print cap × |move|; pre-print cap ≈ erased / |move|) — a bounding identity rather than a computation recipe — whereas skill 1's methodology prefers shares-outstanding × Δprice (sourced from the 10-Q cover) and explicitly labels the percentage method an approximation.
Evidence	File content (Verified).
Impact	Rendering noise in a reference file, and an inverted priority on market-cap arithmetic relative to the sibling skill's stronger doctrine.
Recommended fix	Repair the fence; replace the circular pair with the shares × Δprice formula and keep the percentage method as the labeled cross-check.
Confidence	Verified.

L-4 · Placeholder scan reads script and style blocks (Low)

Table 17: Finding L-4 — Placeholder scan runs on raw HTML including script and style blocks

Field	Content
Location	fact_base_audit.py scan_placeholders() — normalizes the raw file, not visible text
Description	The fact-presence checks strip <script>/<style> and tags before matching, but the placeholder scan matches against the full raw HTML, so a CSS class named placeholder-bar or the JavaScript token undefined inside a script triggers failures on content readers never see.
Evidence	Executed (Verified): a probe slide whose visible text was clean produced “PLACEHOLDER … matches /placeholder/” and “matches /\bundefined\b/” from its style and script blocks. A related dead-code observation: in checks mode, covered.update() adds needle strings in a format the candidate suppressor never consults.
Impact	False positives on code-bearing decks; the scan errs safe but noisy.
Recommended fix	Run the placeholder patterns against visible_text() output only; remove or repair the dead covered.update() in checks mode.
Confidence	Verified (executed).

L-5 · Evidence ledger cites a nonexistent phase (Low)

Table 18: Finding L-5 — Evidence-ledger cites a phase that does not exist

Field	Content
Location	bigtech-equity-research/templates/evidence-ledger.md, Table 2 (“Forward P/E vs band | 11”)
Description	The Data-Gap ledger's Phase column cites “11” for the forward-P/E gap, but skill 3's methodology defines seven phases; 11 is a deliverable section number in report-structure.md, not a phase.
Evidence	File comparison across the three skill-3 files (Verified).
Impact	Minor cross-reference confusion for the executing agent at exactly the moment it is deciding what to re-pull.
Recommended fix	Relabel the column or the value (“Deliverable section 11”).
Confidence	Verified.

L-6 · Embedded sell-side specifics unverifiable (Low)

Table 19: Finding L-6 — Embedded sell-side specifics could not be verified externally

Field	Content
Location	analyzing-us-bigtech-stocks/references/methodology.md — Citizens JMP downgrade, Wedbush PT $445→$387, FactSet consensus $426.20→$404.63 across 61 analysts, BofA Search-at-~10x comparison
Description	The worked example asserts specific named analyst actions and a consensus price-target path as facts from the October 2026 engagement. This audit could not verify them externally within its time box; the surrounding fundamentals in the same worked example (close, FCF, EPS, OCF, Cloud, macro prints) did verify, so these are unverified-by-audit rather than presumed false.
Evidence	External searches returned the event coverage but not these specific action details (Reasoned / Assumed).
Impact	If any specific is wrong, the skill teaches fabrication by example; if all are right, they rot silently as the example ages.
Recommended fix	Move worked-example specifics into a dated, source-linked appendix (URL + retrieval date each), or mark them illustrative; the fact-base schema the skill itself defines already models the right pattern.
Confidence	Assumed (could not be verified or falsified).

3.4 Informational observations

I-1 · No MNPI / Reg FD statement (Informational)

Table 20: Finding I-1 — No MNPI / Regulation FD statement in any of the three skills

Field	Content
Location	All three SKILL.md files
Description	All three skills effectively restrict sourcing to public materials through their tier hierarchies, but none states the compliance principle that only public information is used and material non-public information is out of scope.
Evidence	Full-text reads (Verified — absence).
Impact	A research-controls review under the Wall Street framing of the engagement brief would require the statement, even for an agent tool.
Recommended fix	One sentence in each SKILL.md (“public sources only; if you are in possession of material non-public information, stop”), ideally in the shared standard file proposed under M-3.
Confidence	Verified (absence).

I-2 · No explicit GAAP-to-non-GAAP phase (Informational)

Table 21: Finding I-2 — Skill 3 lacks an explicit GAAP-to-non-GAAP reconciliation phase

Field	Content
Location	bigtech-equity-research/reference/methodology.md (Phases 1–7) and templates/report-structure.md
Description	One-time items appear in skill 3 only inside the Phase 7 waterfall; there is no bridge step. Skills 1 and 2 mandate bridges — skill 2's cash/non-cash × recurring/one-time × dilution classification is the strongest treatment of the three, and skill 1 requires the ex-item arithmetic on the slide.
Evidence	Methodology comparison (Verified).
Impact	An Item 10(e)-style reviewer would ask where the reconciliation lives for skill 3 outputs.
Recommended fix	Add a bridge requirement to Phase 4 or a dedicated section in the report template.
Confidence	Verified.

I-3 · Consensus-vendor attribution varies (Informational)

Table 22: Finding I-3 — Consensus-vendor attribution varies

Field	Content
Location	Skill 1 methodology §2.3 (requires naming the outlet); skill 3 formulas.md worked examples (bare “consensus”)
Description	The audit's own external checks illustrate the stakes: Yahoo's S&P-analyst consensus for Alphabet Q3 2026 revenue was $127.9B (“in line”) while the skills cite $128.2B/$128.22B (miss) — plausibly different vendors, materially different surprise framing. Skill 1 already requires outlet attribution; skill 3's examples do not model it.
Evidence	Yahoo Finance versus the two skills' figures (Verified).
Impact	Unattributed consensus is unreproducible consensus — the first thing a research-controls reviewer asks.
Recommended fix	Require vendor + as-of date on every consensus figure; make skill 3's worked examples carry the attribution skill 1 already mandates.
Confidence	Verified.

4. Skill-by-Skill Assessment

4.1 analyzing-us-bigtech-stocks — RCA deck with audit gates (rubric 4.45 / 5)

Skill 1 is an eleven-phase pipeline (Phases 0–10) that turns a price question into a 12–16-page HTML investment-committee deck on a fixed 1280×720 canvas, styled by a documented “Midnight IC” design system (deep navy, ivory serif, gold accents, monospaced numerals — explicitly referenced to Goldman Sachs / BlackRock IC-memo aesthetics). Its spine is a JSON fact base governed by a written schema: every number carries an ID, category, unit, as-of period, source tier, source name, URL, retrieval timestamp, confidence label, cross-checks, and the hypothesis IDs it evidences. Phases 3–4 force a hypothesis tree of 5–8 competing explanations with falsifiable predictions and at least one explicitly EXCLUDED verdict; Phase 6 mandates the GAAP-to-adjusted bridge with one-off inventory; Phase 7 separates amplifiers (depth) from triggers (direction) and requires a sourced counter-case page; Phase 8 bans price targets; Phase 10 runs the two executable audits before delivery.

Verified strengths. The source-tier table is the most institutional element in the repository: T3 aggregators are barred from originating fundamental claims outright — a rule human analysts routinely break. The T+1-close reaction doctrine (“the after-hours print is thin; the close is what clears real size”) preempts the most common post-earnings analytical error.

The transcript protocol requires verbatim capture of one-off explanations with hedging language intact, and its worked illustration (a $5.2B EC fine inside FCF converting a "cash-flow collapse" into an in-line quarter ex-item) is exactly the craft a senior analyst would teach. The fact base is append-only during an engagement, conflicts are recorded rather than silently resolved, rankings must be sourced or dropped, and the data cutoff is stamped on the cover with an explicit refresh duty. The external verification results were strong: GOOG close $282.32 (−12.29%, −$39.57) confirmed exactly by Bloomberg; FCF −$9.1B and TTM FCF $19.7B confirmed in the earnings-call transcript language; adjusted EPS $2.67 and net income path confirmed by Reuters; OCF $42.3B, Cloud $30.7B (+103%), PCE 3.6% from 3.4%, and the "$482B — worst day ever, second-largest wipeout behind Nvidia Jan 2025" framing all confirmed via Yahoo Finance, Bloomberg and WSJ coverage.

Headline gaps. The numeric audit gate is evadable by digit-substring coincidence (H-4 — verified by execution); the render audit assumes viewport-equals-slide-origin without mandating a body reset (M-1 — verified by probe); the GOOGL cross-class figure conflicts with skill 3's and with accessible public prints (H-1); the placeholder scan matches raw HTML (L-4); and the Playwright prerequisite is undocumented (M-7). fact_base_audit.py itself is genuinely good engineering: exit codes 0/1/2, a JSON report option, an asserting self-test, and a reverse-check that reports slide numbers absent from the fact base as review candidates rather than automatic failures — a smart severity calibration most homegrown QA lacks.

Assessment. Conditional pass at the stated bar. Methodology, sourcing discipline and quality-assurance design are institutional; the gaps are fixable control weaknesses, and the four P0 items in Section 7 close most of the distance.

4.2 big-tech-earnings-postmortem — single-file methodology (rubric 2.85 / 5)

Skill 2 is a nine-step post-earnings workflow in one 283-line file: pin the event numerically (session, point and percentage move, market-cap change, historical rank verified in ≥2 sources, pre-event positioning, sector/session context isolating abnormal return); build a scorecard against consensus across top line, profitability, cash generation, guidance and balance sheet; rank root causes on two axes — durability (mechanical/temporary vs structural) and evidentiary basis (disclosed / inferred / assumed); establish peer and industry context; collect sell-side actions within 48 hours with faithful rationale attribution; build a risk/catalyst matrix whose every row carries an observable, dated trigger; synthesize as a falsifiable thesis with confidence tags; assemble a verification ledger of 10–15 load-bearing claims; and render a restrained, data-dense HTML deliverable.

Verified strengths. The two-axis driver classification is the best single analytical idea across the three skills — durability × evidentiary basis is precisely how a PM triages driver tables, and the skill operationalizes it. The failure-mode catalog reads like desk experience: "flat attribution" (listing causes without ranking), the "accounting-noise hand-wave" (dismissing a GAAP miss without checking whether dilution or a cash fine sits inside it), narrative-first reasoning, peer omission, false precision, "sell-side ventriloquism" (attributing a rationale an analyst never stated) and confidence inflation. The metric-emphasis table per company type is factually sound (capex/OCF and cloud backlog for hyperscalers; China exposure and services attach for Apple; backlog and gross margin for Nvidia; deliveries and auto margin ex-credits for Tesla), and the cash-flow-first ranking heuristic for big-tech moves is a defensible prior stated as a prior.

Headline gaps. No disclaimer language anywhere in the file (H-2) — at the stated bar this alone is disqualifying until fixed. No enforcement: every gate is prose, with no scripts, no templates, no references (M-5). The sourcing rule ("cross-check each figure across at least two independent outlets") lacks skill 1's tier structure and skill 3's data-gap handling. Confidence labeling is good on paper (four labels, including Unverifiable) but unenforced. Routing overlaps the other two skills (M-6).

Assessment. Does not pass at the stated bar in its current form, despite the strongest prose methodology of the three. The gap is not analytical quality; it is compliance posture and the complete absence of enforcement artifacts. Both are cheap to fix, and the P0/P1 items in Section 7 would lift this skill materially.

4.3 bigtech-equity-research — quantified note with strong epistemics (rubric 3.00 / 5)

Skill 3 is a seven-phase process producing a 13-section self-contained HTML research note, with three operating modes (event post-mortem, multi-quarter trend, pre-event setup). Its hard rules are crisply stated: never fabricate; the delta versus consensus moves price, not the absolute; isolate the idiosyncratic move before calling it company-specific; rank root causes and classify each as structural versus cyclical/accounting; tag every figure; label judgment as judgment; no investment advice. Its quant kit covers beat/miss deltas, abnormal return versus QQQ/XLK, options-implied move versus realized, capex-to-OCF ratio, FCF trajectory, an implied-consensus-OCF shortfall identity, average price-target cuts, and a root-cause waterfall with per-factor confidence tags.

Verified strengths. The anti-fabrication protocol and Data-Gap Ledger are the most sophisticated epistemic devices in the repository: when tools are offline the skill shrinks scope to verified-in-context data, tags every gap Assumed or Unverifiable with explicit bounds, and ledgers the exact source pull needed to close it — rather than backfilling from

memory. The rule that no “executed verification” claim may be made for price targets or forward guidance correctly distinguishes reported third-party actions from observable outcomes. All seven quant_helpers functions executed correctly against their documented targets in this audit (revenue delta −0.64B / −0.50%; EPS delta −0.90 / −29.80%; alpha −11.1pp; capex/OCF 121.4%; average PT cut −13.7%; waterfall sum 12.6) — the arithmetic is sound even where the governance around it is not (M-4). The miss-versus-deceleration distinction in Phase 3 is a genuinely institutional touch most retail analysis misses.

Headline gaps. The segment map misstates ASC 280 reportable segments for Nvidia, Amazon and Apple (H-3, verified against the 10-Ks) — in the very file that declares EDGAR segment tables the source of truth. The worked example conflicts with skill 1's on the same event (H-1). The self-test does not assert, and formulas.md's OCF-shortfall example carries a sign error against the code (M-4). The sourcing doc contradicts the methodology on transcript tiers (M-3). There is no GAAP-to-non-GAAP bridge phase (I-2), and the waterfall mandate lacks skill 1's computability guard (M-2).

Assessment. Does not pass at the stated bar as shipped. The epistemic framework is the best of the three and the quant kit is sound, but a reference file with ASC 280 errors in a skill whose Phase 3 depends on it is the kind of defect that destroys credibility with an institutional reader on first inspection.

5. Cross-Skill Comparison and Scorecard

5.1 Portfolio comparison matrix

The three skills are best understood as one product family with three divergent implementations of the same brief. The matrix below compares them across the dimensions an institutional reviewer would tabulate first.

Table 23: Portfolio comparison of the three skills

Dimension	Skill 1 · analyzing-us-bigtech-stocks	Skill 2 · big-tech-earnings-postmortem	Skill 3 · bigtech-equity-research
Primary deliverable	12–16-page HTML deck + keyboard viewer (“Midnight IC” system, 1280×720)	7-section restrained HTML deliverable	13-section self-contained HTML research note
Artifacts shipped	8 files: SKILL.md, 2 JSON templates, 3 references, 2 scripts	1 file: SKILL.md (283 lines)	8 files: SKILL.md, 4 references, 2 templates, 1 script

Dimension	Skill 1 · analyzing-us-bigtech-stocks	Skill 2 · big-tech-earnings-postmortem	Skill 3 · bigtech-equity-research
Enforcement gates	2 executable audits + Definition-of-Done checklist	Definition-of-Done checklist only — no scripts	Definition-of-Done checklist + non-asserting quant helper
Confidence taxonomy	3 labels; Verified = ≥1 primary source, decision-critical double-sourced	4 labels (adds Unverifiable); Verified = executed and observed	4 labels; Verified = ≥2 independent primary sources, consistent to precision
Attribution doctrine	Ordinal ranks; percentages only if computable from sourced data	Ordinal ranks only; percentages are “false precision”	Quantified waterfall summing to the realized move, weights tagged Reasoned
Segment reference data	None embedded — process guidance only	None embedded	Embedded map; 3 of 7 companies misstate ASC 280 segments
Worked example	Alphabet Q3 2026; externally verified in most respects (Bloomberg-exact GOOG close)	None	Alphabet Q3 2026; conflicts with skill 1 (GOOGL move, consensus FCF)
Disclaimer present	Yes — SKILL.md, methodology, templates	No — zero occurrences	Yes — description, hard rules, sourcing doc
License declared	No	Apache 2.0 (frontmatter)	No
Rubric average (0–5)	4.45	2.85	3.00

5.2 Doctrinal conflicts

Beyond packaging, the three skills disagree on doctrine — the rules an executing agent would follow differently depending on which skill fires. These conflicts are the strongest argument for the shared-standard remediation in Section 7.

Table 24: Doctrinal positions in direct conflict

Question	Skill 1	Skill 2	Skill 3
May the move be allocated in percentages?	Only if computable from sourced data; otherwise state qualitative	Never — use ordinal ranks	Yes — waterfall must sum to the realized move
What earns the Verified label?	Observed in ≥1 named primary source; decision-critical cross-checked in ≥2	Executed/checked and observed directly	≥2 independent primary/near-primary sources, consistent to precision
Where do Motley Fool transcripts sit?	Tier 1 (primary), alongside EDGAR and company IR	Not specified	Preference tier 2 in methodology; tier 6 aggregator in sourcing doc (internal conflict)
Market-cap delta method	Shares outstanding × Δprice (10-Q cover), percentage method labeled approximation	Not specified	Pre-print market cap × |move| as the stated formulas
What happens when tools are offline?	Data cutoff restated; refresh duty before delivery	Not specified	Anti-fabrication protocol + Data-Gap Ledger with exact pulls
Deliverable format	Fixed-canvas deck + viewer	HTML slide sequence	Single-file semantic HTML note (WCAG-AA, print-friendly)

5.3 Institutional-readiness scorecard

The scorecard below expresses the audit's conclusions on ten dimensions, scored 0–5 against fixed anchors: 0 = absent; 1 = mentioned only; 2 = partial or inconsistent; 3 = present with material gaps; 4 = present and enforced with minor gaps; 5 = institutional grade. The scores are the auditor's reasoned judgments anchored to the findings in Section 3 and the verification ledger in Section 6 — they are assessments, not measurements, and the anchors are stated so the reader can recalibrate.

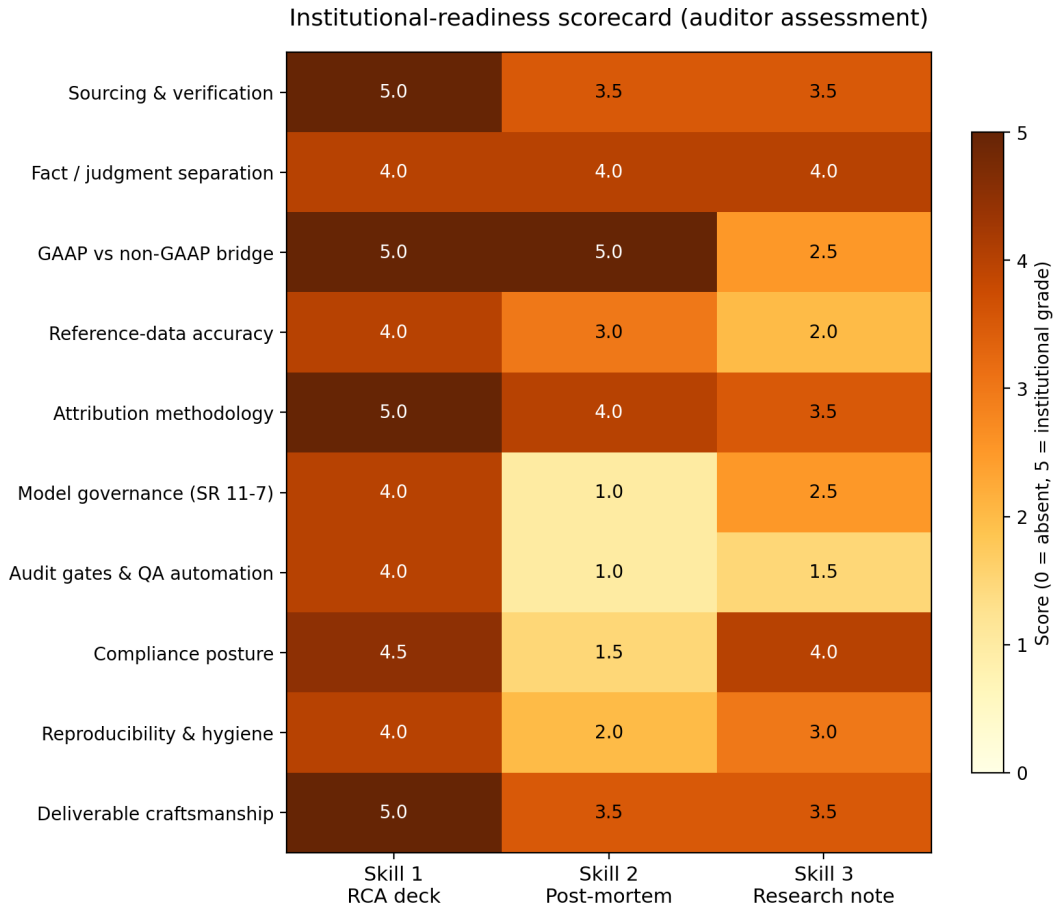


Figure 1: Institutional-readiness scorecard, ten dimensions × three skills (auditor assessment, 0–5)

Table 25: Scorecard with per-dimension basis

Dimension	Skill 1	Skill 2	Skill 3	Basis (finding refs)
Sourcing & verification discipline	5.0	3.5	3.5	Tier system vs two-outlet rule vs hierarchy with internal contradiction (M-3)
Fact / judgment separation & confidence labels	4.0	4.0	4.0	All three label rigorously; taxonomies differ (M-3)
GAAP vs non-GAAP reconciliation	5.0	5.0	2.5	Skill 2's cash/non-cash × recurring/one-time split strongest; skill 3 has no bridge phase (I-2)
Reference-data accuracy (segments, examples)	4.0	3.0	2.0	Skill 1 verified externally with one disputed figure (H-1); skill 3's segment map in error (H-3); skill 2 embeds none

Dimension	Skill 1	Skill 2	Skill 3	Basis (finding refs)
Attribution methodology rigor	5.0	4.0	3.5	Skill 1's conditional rule is the defensible canon; skill 3 mandates without the guard (M-2)
Model governance (SR 11-7 analogy)	4.0	1.0	2.5	Asserting self-test vs none vs non-asserting test and sign error (M-4, M-5)
Audit gates & QA automation	4.0	1.0	1.5	Two real gates (one evadable, H-4) vs prose-only vs checklist (M-5)
Compliance posture (disclaimer, MNPI, licensing)	4.5	1.5	4.0	Disclaimer present in 1 and 3, absent in 2 (H-2); MNPI statement absent in all (I-1); licensing (L-2)
Reproducibility & engineering hygiene	4.0	2.0	3.0	Deterministic helpers and JSON audits vs environment-coupled Playwright (M-7), packaging drift (L-1)
Deliverable craftsmanship	5.0	3.5	3.5	Documented design system with overflow budget; two deliverable specs unspecified to that depth
Average	4.45	2.85	3.00	—

6. Verification Ledger

Per the audit protocol, the ledger records what was checked, how, and the observed result. Commands were executed on 2026-10-30 against the clone at HEAD 48cd3f97; external sources were consulted the same day. Exit codes are reported where meaningful.

Table 26: Verification ledger (19 checks)

#	Check	Method	Result	Confidence
1	Repository cloned at stated HEAD	git clone --depth 1; git log -1	48cd3f97 · 2026-10-30 13:15 +0800 · “update skills”	Verified
2	File inventory of the three skills	find + wc -l	17 files · 2,078 lines; skill 2 = single file	Verified

#	Check	Method	Result	Confidence
3	fact_base_audit.py self-test	python3 …/fact_base_audit.py --selftest	PASS (exit 0); detects missing token, BADREF page, placeholder	Verified
4	Numeric-audit evasion (substring false-pass)	Custom 4-case harness against the script	82.3 vs “$282.32” → PASS exit 0; 12.29 vs “−112.29%” → PASS exit 0; controls correct	Verified
5	Checks-mode and scan-only mode	Probe slide + checks.json	Missing needle flagged exit 1; placeholder scan flags TODO/placeholder	Verified
6	Placeholder scan scope (raw HTML)	Probe slide with clean visible text + CSS class / JS token	False positives on /placeholder/ and /\bundefined\b/ from code blocks	Verified
7	quant_helpers.py correctness	python3 quant_helpers.py; hand-checked identities	All 7 outputs match documented targets; OCF shortfall prints +10.10B vs formulas.md −10.10B	Verified
8	overflow_audit.mjs behavior	node --check; usage run; probe slide with true overflow + data-bleed element	True overflow flagged (exit 1); body-margin false positive on h1/p (right=1288); Playwright resolved via hardcoded path	Verified
9	GOOG October 29 close	Bloomberg quote page (GOOG:US)	$282.32 · −$39.57 · −12.29% as of 5:20 PM EDT 10/29/26 — matches skill 1 exactly	Verified
10	Prior closes (October 28)	Yahoo Finance pre-event article ticker strip	GOOG $321.89 (+1.60%) · GOOGL $324.89 (+1.34%)	Verified

#	Check	Method	Result	Confidence
11	Alphabet Q3 2026 fundamentals	Yahoo Finance, WSJ, Reuters, earnings-call transcript	Revenue $127.6B (consensus $127.9B Yahoo) · GAAP EPS $2.12 · adj. EPS $2.67 · OCF $42.3B · FCF −$9.1B · TTM FCF $19.7B · Cloud $30.7B (+103%)	Verified
12	Event superlatives	Yahoo, Bloomberg, WSJ headlines	Worst day ever since 2004 IPO; ≈$482B erased (Yahoo), ~$500B (WSJ); second-largest one-day wipeout for a US-listed stock (Bloomberg)	Verified
13	Nvidia reportable segments	SEC EDGAR FY2026 10-K (nvda-20260125.htm)	Two segments: Compute & Networking, Graphics — segment-maps.md in error	Verified
14	Amazon reportable segments	SEC EDGAR FY2025 10-K (amzn-20251231.htm)	Three segments: North America, International, AWS; advertising is a net-sales line — segment-maps.md in error	Verified
15	Apple reportable segments	10-K structure via multiple secondary confirmations	Five geographic segments; product lines are revenue disaggregation — segment-maps.md in error	Verified
16	GOOGL October 29 close adjudication	Yahoo article/ticker prints; Bloomberg GOOGL page (blocked)	Prints −12.78% / −12.79% (283.35–283.37); matches neither skill (−11.71% / −12.6%); official close unconfirmed	Reasoned

#	Check	Method	Result	Confidence
17	Disclaimer presence	Case-insensitive grep across the three skills	Skills 1 and 3: multiple hits; skill 2: zero	Verified
18	Licensing and catalog indexing	Filesystem check; case-insensitive grep of skills-catalog.md and skills-inventory.md	No root LICENSE; no catalog/inventory entry for any of the three skills	Verified
19	Impact arithmetic	Python computation	12.15B × $39.57 = $480.8B ≈ press $482B; GOOGL move gap 0.89pp ≈ $35.1B on ≈$3.95T; avg PT cut −13.7%; rev delta −0.50%	Verified

7. Remediation Roadmap

The roadmap is sequenced by the standard applied throughout: fix what blocks the stated bar first, then governance, then hygiene. Effort estimates assume the repository maintainer executes with the audit's evidence in hand; every P0 item is independently verifiable after the fix using the same checks recorded in Section 6.

7.1 P0 — ship blockers at the stated bar (≈1 day combined)

Table 27: P0 remediation items

#	Action	Closes	Effort	Verification after fix
1	Restructure segment-maps.md into “reportable segments” vs “revenue disaggregation” with per-company vintages and refresh cadence	H-3	30 min	Diff against NVDA FY2026 / AMZN FY2025 / AAPL 10-K segment tables
2	Add disclaimer + public-sources statement to skill 2's quality gates and Step 9 design constraints	H-2, I-1	15 min	Grep for disclaimer language returns hits in all three skills
3	Boundary-anchor token_in() in fact_base_audit.py; add the two evasion cases to --selftest	H-4	1 h	The Section 6 harness returns MISSING + exit 1 for both substring cases

#	Action	Closes	Effort	Verification after fix
4	Reconcile worked examples into one canonical, dated, sourced file; correct or source the GOOGL figure and the consensus figures	H-1	2 h	Cross-skill diff of shared datapoints returns zero conflicts; GOOGL figure matches an exchange close record

7.2 P1 — governance and control quality (≈1 week)

- Extract a shared sourcing / confidence / attribution standard (one taxonomy, one source hierarchy with transcript tiering resolved, one attribution policy) consumed by all three skills; add a repository-level consistency test that diffs shared doctrine and shared datapoints at commit time (M-2, M-3, H-1 recurrence).
- Fix the render audit: overflow relative to the .slide rectangle or a mandated html,body reset in the token sheet; offline-tolerant page-load strategy; document the Playwright prerequisite and remove machine-specific absolute paths (M-1, M-7).
- Convert quant_helpers.py to an asserting self-test; correct the formulas.md OCF-sign example or define the convention in both places; add pytest with zero-division and empty-input cases (M-4).
- Port the fact-base and audit-gate pattern into skill 2 (or delegate to skill 1's scripts), reuse skill 3's evidence-ledger template, and demarcate the three skills' descriptions with cross-references; register all three in skills-catalog.md (M-5, M-6).

7.3 P2 — hygiene (as convenient)

- Standardize folder conventions and frontmatter across the three skills (L-1); add a root LICENSE and per-skill license fields (L-2).
- Repair the formulas.md code fence and replace the circular market-cap pair with shares × Δprice as primary (L-3); scope the placeholder scan to visible text (L-4); relabel the evidence-ledger phase reference (L-5).
- Move the worked example's analyst-action specifics into a dated, source-linked appendix, or mark them illustrative (L-6); add a GAAP-bridge phase to skill 3 (I-2); require consensus-vendor attribution in every worked example (I-3).

Sequencing note: items 1–4 of P0 are independent of each other and can land as four atomic commits; the P1 shared-standard work should follow the P0 reconciliation so the canonical example file becomes the first consumer of the shared standard rather than a fourth divergent copy.

8. Appendix A — File Inventory

The seventeen in-scope files with line counts and roles, as read in full during this audit (HEAD 48cd3f97).

Table 28: Audited file inventory

File	Lines	Role
analyzing-us-bigtech-stocks/SKILL.md	238	Entry point: triggers, confidence system, 11-phase pipeline, bundled-resources map
…/assets/fact_base.template.json	75	Fact-base skeleton with worked example patterns
…/assets/slides_brief.template.json	32	15-page deck-plan skeleton with per-slide task briefs
…/references/fact-base-schema.md	153	Field definitions, confidence rules, conflicts, worked examples
…/references/methodology.md	355	Full phase-by-phase playbook, source tiers, pitfall catalog, compliance notes
…/references/report-design.md	135	Midnight IC tokens, component library, 15-page archetype map, overflow budget
…/scripts/fact_base_audit.py	228	Numeric + placeholder audit (modes: fact-base, checks, scan, selftest)
…/scripts/overflow_audit.mjs	126	Playwright render/overflow audit for 1280×720 decks
big-tech-earnings-postmortem/SKILL.md	283	Single-file 9-step post-earnings RCA methodology
bigtech-equity-research/SKILL.md	111	Entry point: hard rules, modes, workflow, anti-fabrication protocol
…/reference/formulas.md	73	Quant formulas with worked examples
…/reference/methodology.md	109	Seven-phase research process
…/reference/segment-maps.md	20	Reportable segments per company (subject of H-3)

File	Lines	Role
…/reference/sourcing-and-confidence.md	35	Source hierarchy, confidence taxonomy, anti-hallucination rules, disclaimer
…/scripts/quant_helpers.py	55	Deterministic delta/alpha/waterfall math (subject of M-4)
…/templates/evidence-ledger.md	21	Evidence + Data-Gap ledger format
…/templates/report-structure.md	29	13-section deliverable spec

9. Appendix B — External Sources Consulted

External sources consulted during verification, with what each was used to establish. All retrievals occurred 2026-10-30; quotes are as retrieved. Paywalled or robot-blocked pages are marked and their accessible content described.

Table 29: External sources and what they verified

Source	Used to establish	Reference
Bloomberg — GOOG:US quote page	GOOG official close $282.32, −$39.57, −12.29% (as of 5:20 PM EDT 2026-10-29); “worst drop on record; second biggest-ever market-value wipeout”	bloomberg.com/quote/GOOG:US (GOOGL page robot-blocked)
Yahoo Finance — event coverage	$482B market-value loss headline; GOOG/GOOGL ticker prints 281.14/283.37 (−12.66%/−12.78%); intraday article 281.09/283.35; revenue $127.6B vs $127.9B S&P-analyst consensus; GAAP EPS $2.12; ad revenue short of estimates with Cloud beat at $30.7B; PCE 3.6% from 3.4%; Fed hold	finance.yahoo.com — Alphabet Q3 2026 coverage set
Yahoo Finance — pre-event article ticker	October 28 closes: GOOG $321.89 (+1.60%), GOOGL $324.89 (+1.34%)	finance.yahoo.com — pre-earnings preview page
Reuters	Net income to common fell 25% to $26.17B; adjusted EPS $2.67 missed Wall Street	reuters.com — “Alphabet stock sinks as search woes cause revenue miss”

Source	Used to establish	Reference
WSJ	Worst day on record; roughly $500B erased; $127.6B sales narrowly missing expectations; Cloud $30.7B beat	wsj.com — Alphabet selloff coverage
Earnings-call transcript (via Yahoo Finance)	“We had negative free cash flow of $9.1 billion in the third quarter, driven by our investments in CapEx”; FCF $19.7B TTM; OCF $42.3B	Alphabet Q3 FY2026 earnings call transcript
SEC EDGAR — NVIDIA FY2026 10-K	Two reportable segments (Compute & Networking; Graphics) with Data Center and Automotive inside C&N — segment-maps.md error confirmed	sec.gov/Archives/edgar/data/1045810/000104581026000021/nvda-20260125.htm
SEC EDGAR — Amazon FY2025 10-K	Three reportable segments (North America, International, AWS); advertising as net-sales driver, not a segment	sec.gov/Archives/edgar/data/1018724/000101872426000004/amzn-20251231.htm
Apple 10-K structure (secondary confirmations)	Five geographic reportable segments; FY2025 net sales Americas $178.4B, Europe $111.0B, Greater China $64.4B	stock-analysis-on.net; secfilingdata.com; academic case analyses
Amazon IR — Q4 2025 results release	Segment tables for North America / International / AWS only	ir.aboutamazon.com — fourth-quarter results
Business Insider / mlq.ai / Barron's headlines	Corroborating event framing, analyst-reaction timing, sector tape	businessinsider.com; mlq.ai; barrons.com
Stooq daily CSV (attempted)	Independent close adjudication — blocked by JavaScript verification wall; noted as a limitation	stooq.com (inaccessible to fetcher)